

Khaing Myal Htike

AI & BACKEND DEVELOPER

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Professional Summary

Backend developer with three years of experience building and running production systems for AI products. Built the Python backend behind Magick Box, an AI creative app on iOS, Android and the web, from the API and data model through payments, real-time chat and Kubernetes deployment, and more recently the Go API and Flutter app for Magick Whisper. Works across the whole backend stack: REST API design, relational and document databases, caching and queues, authentication and security, testing and CI/CD, containers and cloud. Prefers clean, well-tested code and plain explanations.

Work Experience

General Magick Industries | AI & Backend Developer

Mar 2024 – Present

Bangkok, Thailand

Magick Box, AI creative platform on iOS, Android and the web

- Built the backend in Python 3.12 with FastAPI and Pydantic: around 250 REST endpoints across 21 feature packages.
- Structured the codebase as feature-sliced Clean Architecture with interface-based repositories and services and an async dependency-injection layer.
- Added centralised exception handling, request middleware, a split public and admin OpenAPI surface, and graceful shutdown with per-resource timeouts.
- Designed the MongoDB data model: around 58 collections with indexes and aggregation pipelines.
- Wrote the feed ranker, blending time-decayed vote score with follow, tag and content-type affinity.
- Used Redis for response and user caching with TTL invalidation, a sliding-window rate limiter with per-minute and per-day windows, and Socket.IO pub/sub across pods.
- Integrated text, image, video, music and podcast generation across OpenAI, Google Gemini and Veo, Stability, Kling, Suno and Flux behind a provider abstraction with credit-based model tiers.
- Ran long-running generation on Celery workers with late acknowledgements, exponential backoff and time limits, with FFmpeg for podcast stitching and video conversion.
- Built Mini Cinema, a multi-shot AI film pipeline that streams reasoning, approval gates and renders to the client as typed content blocks.
- Built real-time chat over Socket.IO with token streaming, per-user chat locks and a structured content-block message schema.
- Built group chat mixing human members with AI personas: invitations, admin roles, @mentions, and watermark-based unread tracking with per-user notification fan-out.
- Implemented authentication with device-bound JWT access and refresh tokens, absolute session expiry, Argon2 hashing with legacy-hash migration, email OTP, and Google and Apple sign-in.
- Implemented monetisation with Stripe Checkout and RevenueCat: signature-verified webhooks, an idempotent credit ledger, subscription reconciliation, and AdMob server-side verification for rewarded ads.
- Added a quest and level system for gamification, content moderation, and admin tooling.
- Wrote 181 pytest tests covering chat orchestration, feed ranking, the payments lifecycle and session authorisation.
- Packaged a multi-stage Docker image with uv that serves both the web process and the Celery worker; deployed to Kubernetes on AWS with Helm charts, Istio mTLS and retries, and horizontal autoscaling.
- Set up GitHub Actions workflows for tests, build, release and DAST security scanning.

Magick Whisper, dog-training companion for the Guardian collar (Aug 2026 – present)

- Wrote the Go API with Gin and pgx over PostgreSQL: hand-written SQL and embedded goose migrations with generated columns and partial unique indexes.
- Implemented Google and Apple ID-token verification against cached JWKS, 15-minute access tokens, and rotating hashed refresh tokens.

- Returned RFC 7807 problem responses with per-field validation; added request-ID logging, health and readiness probes, and graceful shutdown.
- Wrote 134 Go tests that run against a real Postgres service in CI, with a check that fails the build if any database test skips; shipped as a distroless Docker image.
- Built the Flutter app with Riverpod and go_router in feature-sliced layers, with Google and Apple sign-in kept optional to meet App Store guideline 5.1.1.
- Added on-device speech recognition with Whisper and Silero VAD through sherpa_onnx in an isolate, an offline session store with retry-safe UUIDv7 uploads, and secure token storage.
- Built light and dark themes backed by WCAG contrast tests, with around 480 widget and unit tests across the app.
- Set up GitHub Actions for analysis and tests and Codemagic for signed iOS and Android releases to TestFlight and Play, plus a separate UI demo build with golden screenshot tests.

Yoma Bank | Full Stack Developer

Oct 2023 – Mar 2024

Yangon, Myanmar

- Built internal compliance and audit tools for the bank's legal team, replacing manual checks with a tracked workflow.
- Designed and maintained REST APIs in Java, with relational schema design and SQL for the reporting datasets behind them.
- Processed data for internal reports and worked in a Git workflow with code reviews.
- Built front-end features in JavaScript, HTML and CSS.

Education

Liverpool John Moores University

2021 – 2024

BSc Computer Network and Infrastructure. Liverpool, United Kingdom.

Skills

Languages: Python, Go, Java, Dart, SQL, JavaScript

Backend: FastAPI, Pydantic, Gin, REST and OpenAPI, WebSockets (Socket.IO), Clean Architecture, dependency injection, async programming

Databases: PostgreSQL (pgx, goose migrations), MongoDB (aggregation pipelines, indexing), Redis

Async, caching and resilience: Celery, Redis caching, sliding-window rate limiting, idempotent webhooks, retries with backoff, FFmpeg

Auth and security: JWT, OAuth and OIDC (Google, Apple), Argon2, webhook signature verification, DAST scanning

AI: LiteLLM, OpenAI, Google Gemini and Veo, Groq, Stability AI, Suno, Kling, RAG, on-device Whisper (sherpa_onnx)

Testing and delivery: pytest, Go testing, Flutter widget tests, GitHub Actions, Codemagic, Docker, Helm, Kubernetes, Istio, AWS, Render

Mobile and payments: Flutter, Riverpod, go_router, Stripe, RevenueCat, AdMob server-side verification

Spoken languages: Burmese (native), English (professional), Thai (conversational)

Projects

Magick Box. AI creative platform for image, video, music and podcast generation, with a social feed, chat and Mini Cinema films. [App Store](#) · [Google Play](#) · [magickbox.ai](#)

Magick Whisper. Reward-based dog-training app for the Guardian collar, in development: Go API on PostgreSQL, Flutter app with on-device speech recognition, and a separate clickable UI demo.